Proofs with Sets

If A, B, and C are sets, then $A \setminus (B \cup C) = (A \setminus B)$ ()
$(L \subset R)$ We have $L = A \setminus (B \cup C)$ adn $R = (A \setminus B) \cap (A \setminus C)$.

Proof. By DeMorgan's Law, $x \notin B$ and $x \notin C$.
Then, $x \in A \setminus B$ and $x \in A \setminus C$.
Assume x$\in R$. Then, $x \in A \setminus B$ and $x \in A \setminus C$
So $x \notin B \cup C$.
It follows that $x \in A$ but $x \notin B \cup C$. □

Suppose A, B, and C are sets. Prove that if $B \subseteq C$, then $A \times B \subset A \times C$.

Proof. Assume that $z \in A \times B$.
Then, $x \in A$ and $y \in B$ such that $z = (x, y)$.
Since $B \subseteq C$ and $y \in B$, it follows that $y \in C$.
So $x \in A$ and $y \in C$.
Then, $(x, y) \in A \times C$.
So $z \in A \times C$. □

For any sets A and B, $\mathbb{P}(A) \cap \mathbb{P}(B) = \mathbb{P}(A \cap B)$

Proof. Assume $X \in P(A) \cap P(B)$.
Then, $X \in P(A)$ and $X \in P(B)$
So, $X \subseteq A$ and $X \subseteq B$.
Then, $X \subseteq A \cap B$.
Then $X \in P(A \cap B)$.
For the converse, we assume $X \in P(A \cap B)$.
Then, $X \subseteq A \cap B$. It follows that $X \subseteq A$ and $X \subseteq B$.
So, $X \in P(A)$ and $X \in P(B)$.
It follows that $X \in P(A) \cap P(B)$. □

$\forall A, B$, prove that $A \subseteq B \iff A \setminus B = \emptyset$.

Proof. Assume $A \setminus B = \emptyset$.

That is, $A \subset B$.

For the converse, assume $A \setminus B \neq \emptyset$. □

Proof. Let $A = \{0, 1, 2\}$, $B = \{2, 4\}$, and $C = \{3, 5\}$. Then, $(B \cup C) = \{2, 3, 4, 5\}$. So $A \setminus (B \cup C) = 0, 1$. Then, $(A \setminus B) = 0, 1$ and $A \setminus C = \{0, 1, 2\}$ □